BATTERY COMPARISON ANALYSIS SYSTEM

Comparative Analysis of the Performance of Lithium-Ion and Lead-Acid Batteries for a Solar System in FUT Minna Engineering Complex

Study Location:
FUT Minna Engineering Complex

Data Source:
Project CSV Dataset

Report Generated:
09 September 2026, 23:49

Automated Analytical Report

1. Introduction

Energy storage is an important component of solar energy systems because batteries provide the means for storing electrical energy generated during periods of solar availability and supplying power when solar generation is insufficient.

This analysis provides a comparative assessment of lithium-ion and lead-acid battery performance using the supplied charging efficiency, discharge capacity and depth of discharge datasets.

The purpose of the application is to provide a consistent computational framework for examining the supplied battery performance data and presenting the results in tables and graphical form.

2. Methodology

The analysis system reads three comma-separated value (CSV) datasets. The datasets represent charging efficiency, discharge capacity and depth of discharge for lithium-ion and lead-acid batteries.

The data are processed using descriptive statistical analysis. For each battery type, the system calculates the number of observations, mean, standard deviation, minimum value, maximum value, range and coefficient of variation where applicable.

The comparative analysis also determines the difference between the lithium-ion and lead-acid values for corresponding observations.

Data integrity note: The report uses only the values contained in the project's CSV files. No additional experimental observations are generated by the software.

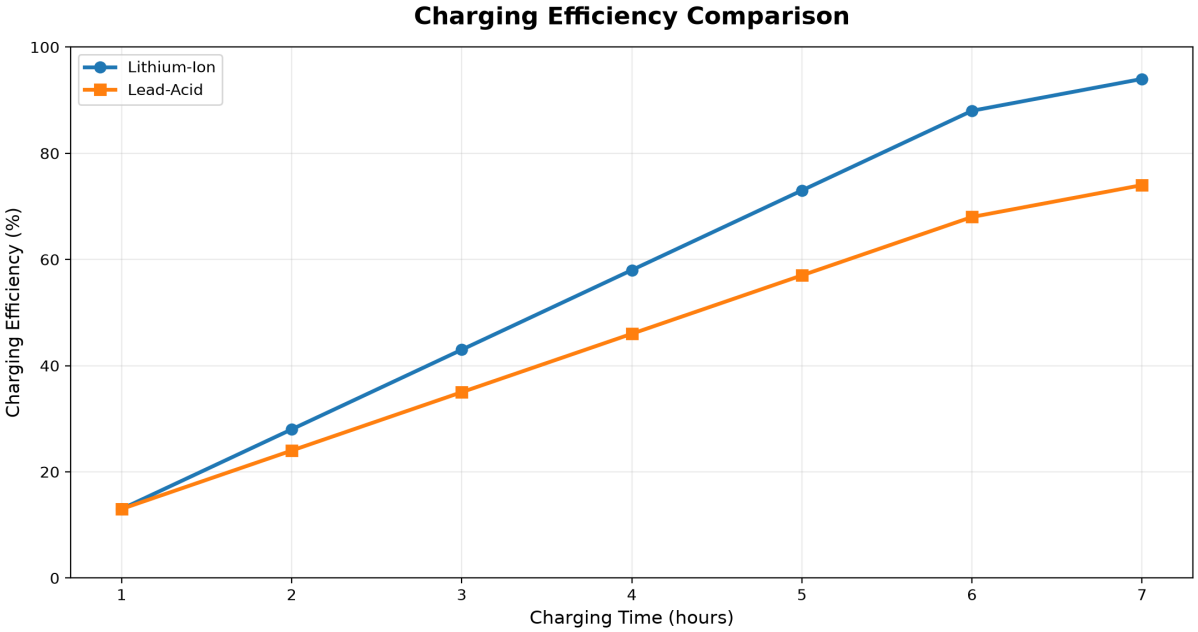
Key Performance Indicators

Performance Indicator	Lithium-Ion	Lead-Acid
Mean Charging Efficiency (%)	56.71	45.29
Mean Discharge Capacity (%)	85.00	70.00
Mean Depth of Discharge (%)	92.14	51.86

3. Charging Efficiency Analysis

The supplied charging-efficiency dataset was examined to determine the central tendency and variation of the recorded values for both battery types.

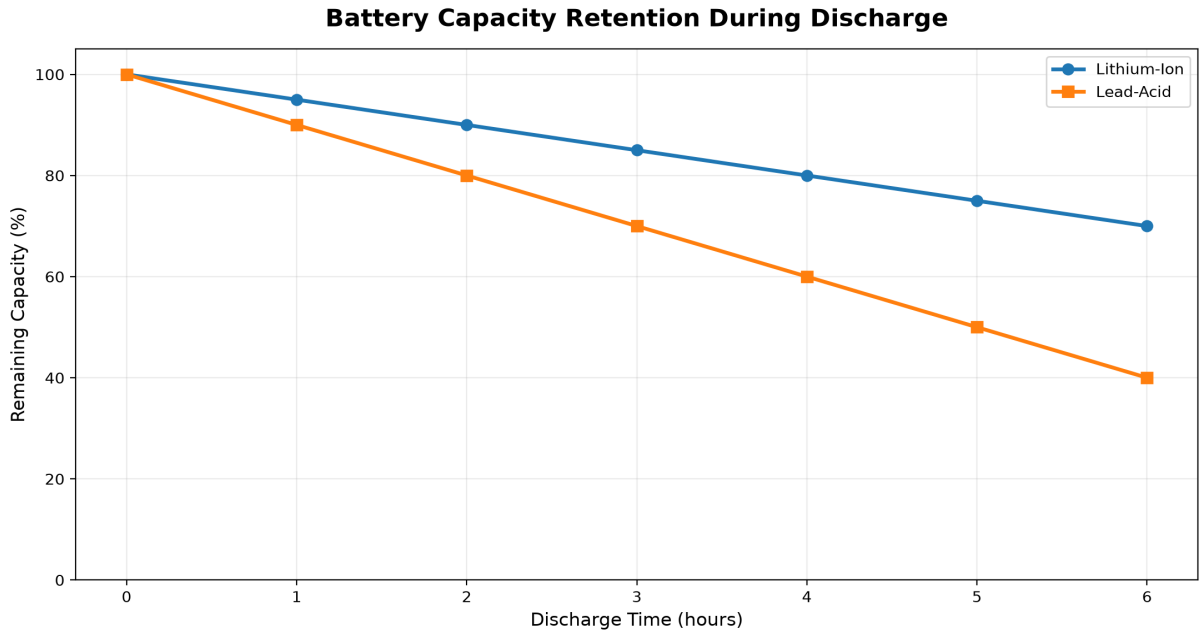
Battery Type	Mean (%)	SD	Minimum (%)	Maximum (%)
Lithium-Ion	56.71	30.44	13.00	94.00
Lead-Acid	45.29	22.65	13.00	74.00


Charging Efficiency Comparison
Lithium-Ion
Lead-Acid
Charging Efficiency (%)
Charging Time (hours)
0
20
40
60
80
100
1
2
3
4
5
6
7

4. Discharge Capacity Analysis

Discharge capacity values are presented as the percentage capacity values supplied in the dataset. The analysis therefore treats these values as percentage-based capacity measurements rather than absolute ampere-hour capacity.

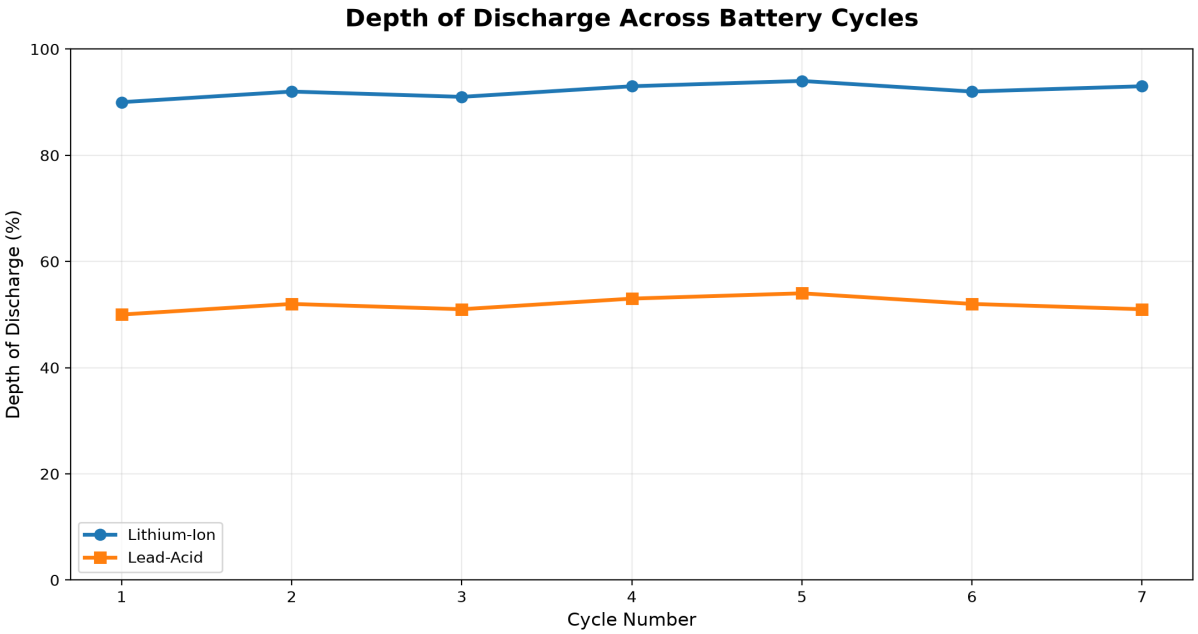
Battery Type	Mean (%)	SD	Minimum (%)	Maximum (%)
Lithium-Ion	85.00	10.80	70.00	100.00
Lead-Acid	70.00	21.60	40.00	100.00



5. Depth of Discharge Analysis

The depth-of-discharge dataset was examined across the supplied battery cycles. The values are reported as percentages in accordance with the supplied data.

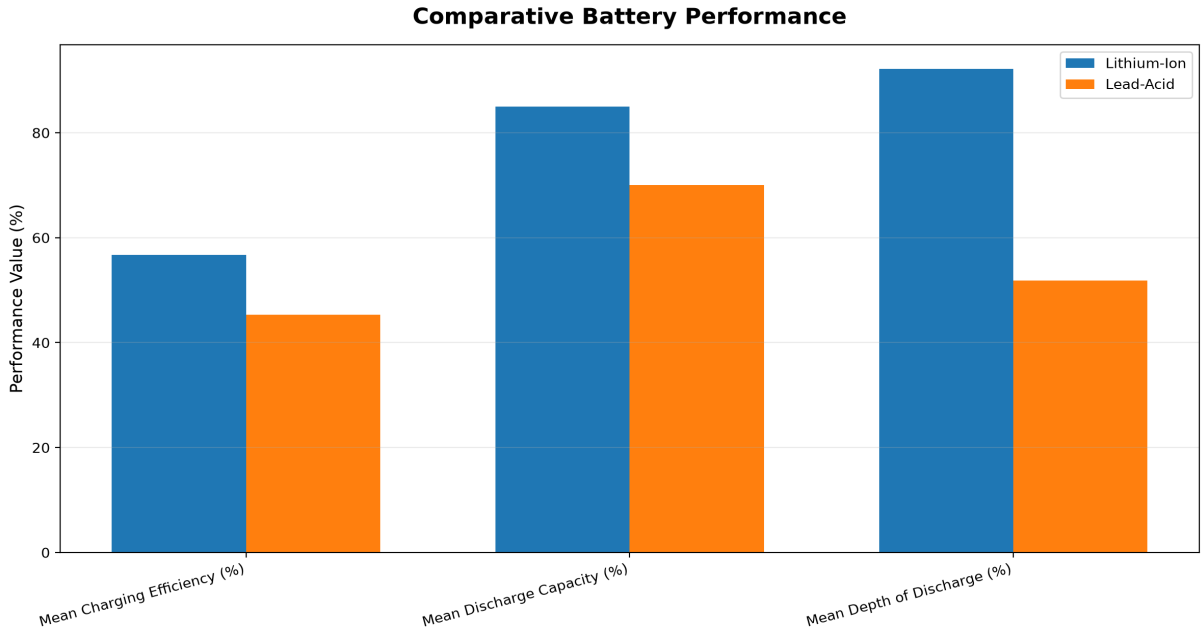
Battery Type	Mean (%)	SD	Minimum (%)	Maximum (%)
Lithium-Ion	92.14	1.35	90.00	94.00
Lead-Acid	51.86	1.35	50.00	54.00



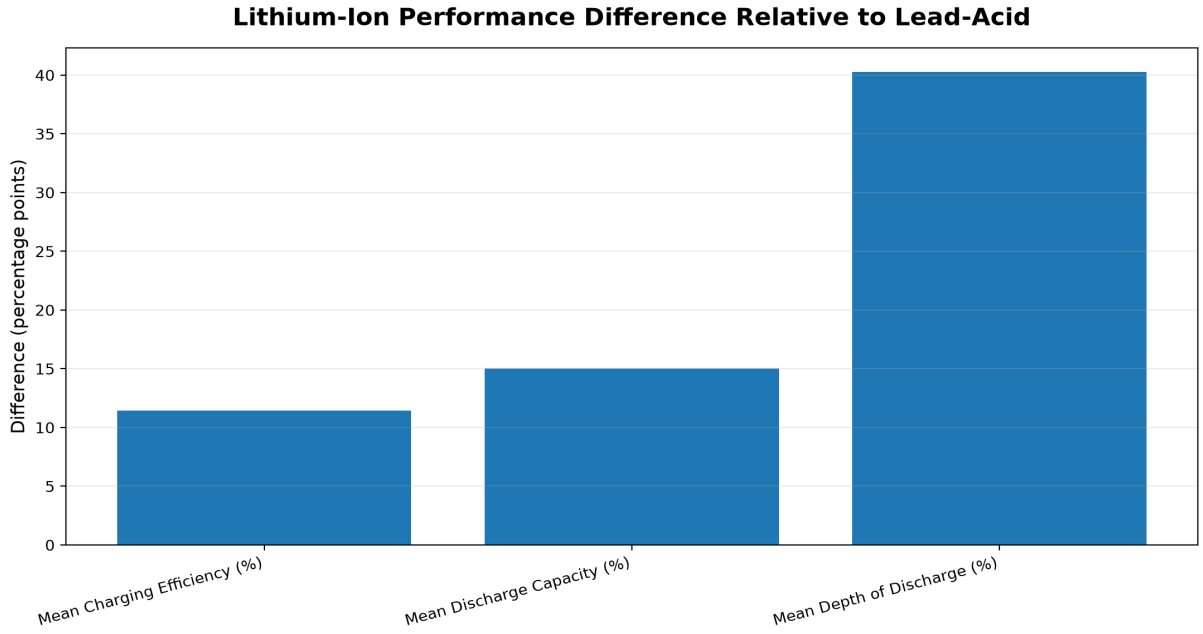
6. Comparative Analysis

The following table summarizes the mean values obtained for the three investigated performance parameters.

Performance Parameter	Lithium-Ion	Lead-Acid	Difference	Percentage Difference
Mean Charging Efficiency (%)	56.71	45.29	11.43	25.24
Mean Discharge Capacity (%)	85.00	70.00	15.00	21.43
Mean Depth of Discharge (%)	92.14	51.86	40.29	77.69



Comparative Difference



7. Discussion of Results

The mean charging-efficiency value for the lithium-ion battery was **56.71%**, compared with **45.29%** for the lead-acid battery. The corresponding difference was 11.43 percentage points.

For the supplied discharge-capacity observations, the mean lithium-ion value was **85.00%**, while the lead-acid value was **70.00%**. The difference between the means was 15.00 percentage points.

The mean depth-of-discharge values were **92.14%** for lithium-ion and **51.86%** for lead-acid, giving a difference of 40.29 percentage points.

Interpretation note: A higher depth of discharge is not automatically equivalent to better overall battery performance. Depth of discharge should be interpreted together with battery life, cycle life, operating conditions and the specific experimental procedure.

The numerical conclusions in this report are limited to the supplied dataset. They should not be presented as measured field performance unless the supplied values correspond to the actual experimental measurements obtained by the researcher.

8. Conclusion

Based on the supplied dataset, the lithium-ion battery recorded a mean charging-efficiency value of 56.71%, while the lead-acid battery recorded 45.29%.

The mean discharge-capacity values were 85.00% for lithium-ion and 70.00% for lead-acid under the supplied observations.

The analysis therefore provides a quantitative basis for comparing the two battery technologies using the selected performance indicators. However, final engineering conclusions should consider the actual experimental conditions, battery specifications and measurement procedures used during the research.